Topic: (Integral of rational expression of polynomial) ⇒ 7.5

Rule 1: If the numerator is derivative of ~~num~~ denominator then we write the antiderivative of the Rational expression as $\ln|\text{denominator}|$.

i.e $\int \frac{g(x)}{f(x)}\,dx = \ln|f(x)| + c$, where $g(x) = f'(x)$ and $f(x) \neq 0$

Rule 2: If $g(x) \neq f'(x)$, then rule 1 will not work. We will follow the following steps to ~~evalua~~ integrate $\frac{g(x)}{f(x)}$, $f(x) \neq 0$.

Step 1: $\frac{g(x)}{f(x)}$ proper ~~of~~ or Improper fraction ?? If it is proper go to the step 2, Otherwise step $\boxed{6}$

How can you check it?

fraction

- Proper: degree $(g(x))$ < degree $(f(x))$
- Improper: degree of $(g(x))$ ≥ degree of $(f(x))$.
- Mixed

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Example :

$$\frac{2x^2+5x}{x^3-6}$$

$2x^2+5x \rightarrow$ degree 2

$x^3-6 \rightarrow$ degree 3

$\downarrow$ proper fraction

$$\frac{x^3-6}{2x^2+5x}$$

$x^3-6 \rightarrow$ degree 3

$2x^2+5x \rightarrow$ degree 2

$\rightarrow$ Improper fraction.

Step 2 : factorize the denominator to write it as the product of prime factors.

Step 3 : For every factor in denominator follow the following list to write the partial fraction decomposition.

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Factor	Partial fraction
Linear $(ax+b)$	$\frac{A}{ax+b}$.
Linear with repeated factors. $(ax+b)^n$	$\frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{C}{(ax+b)^3} + \cdots$ upto $(ax+b)^n$.
Quadratic ax^2+bx+c	$\frac{Ax+B}{(ax^2+bx+c)}$
Qadratic repeated fact $(ax^2+bx+c)^n$	$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{ax^2+bx+c} + \cdots + \frac{A_nx+B_n}{ax^2+bx+c}$

Follow the same pattern for any ordered polynomial and its repeated factors.

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Step 4: Find all arbitrary constants (coeffecients) values.

A, B, C, A_1, B_1, C_1 et

There are several methods for determining the coeffecients for each term.

Step 5: Finally ~~it~~ integrate.

Step 6: For Improper fraction, partial fraction will not be applicable.

To fix the problem, ~~us~~ use long division on the improper fraction.

for an example $\boxed{\dfrac{x^3-3x^2}{x-2}}$ Improper fracti

$$\frac{x^4-5x^3+6x^2-18}{x^3-3x^2}.$$

$$\cancel{x-2 \,\big|\, x^3-3x^2 \quad x^2 \quad x^3 \quad -2x \quad 2x^2 -}$$

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Long Division

$$
\begin{array}{r|l}
 & x - 2 \\
\hline
x^3 - 3x^2 & x^4 - 5x^3 + 6x^2 - 18 \\
 & \underline{x^4 - 3x^3} \\
 & \quad -2x^3 + 6x^2 - 18 \\
 & \quad \underline{-2x^3 + 6x^2} \\
 & \qquad\qquad -18
\end{array}
$$

$$\therefore \frac{x^4 - 5x^3 + 6x^2 - 18}{x^3 - 3x^2} = x - 2 + \frac{-18}{x^3 - 3x^2}$$

$$= \underbrace{(x-2)}_{\text{round}} + \underbrace{\frac{-18}{x^3 - 3x^2}}_{\text{proper fraction}}$$

$+$

Mixed fraction.

Step 7:- follow step 2 to step 5.

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